

B.Tech. Degree III Semester Supplementary Examination
April 2018

ME 15-1303 MECHANICS OF SOLIDS
 (2015 Scheme)

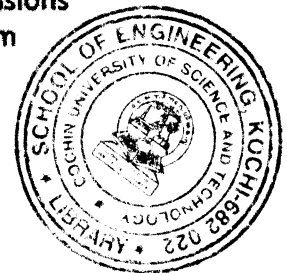
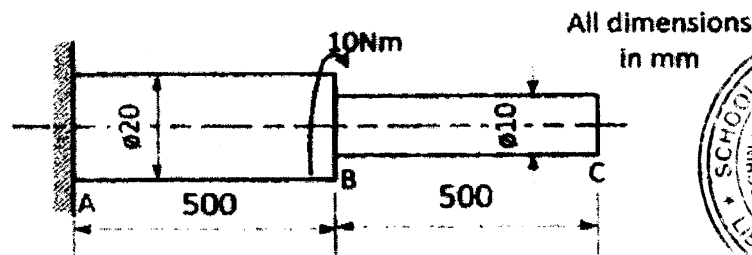
Time : 3 Hours

Maximum Marks : 60

PART A
 (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Draw the stress-strain curve of cast iron. What property of cast iron did you use to draw its stress strain curve?
- (b) Draw Mohr's diagram to represent biaxial stresses, where $\sigma_y = -\sigma_x$. Find out the maximum shear stress (τ_{xy}) and the plane on which it acts.
- (c) Define hoop stress. Derive an expression for hoop stress for a cylindrical pressure vessel (thin cylinder) with radius r and thickness t .
- (d) For the stepped bar shown in the figure find the angle of twist at C in degrees, $G = 77 \text{ GPa}$.



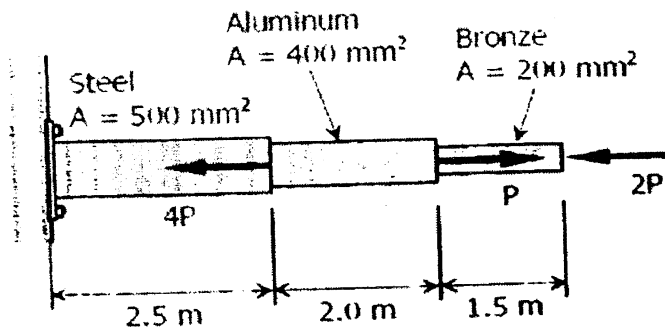
- (e) Draw the bending moment diagram for a simply supported beam of 1000 mm length with a counter clockwise (CCW) moment of 1 kNm at 250 mm from the left end and a clockwise (CW) moment of 1 kNm at 750 mm from the left end.
- (f) What is section modulus? Find out the section modulus of a pipe with outer diameter d_1 and inner diameter d_2 .
- (g) List and briefly explain various failure theories. Which theory is best suited for the failure analysis of brittle materials?
- (h) Find out the maximum deflection of a cantilever beam of length L and constant EI with a moment M applied at the free end.
- (i) Explain how a uniformly varying load from 0 to W over a length of $\frac{L}{2}$ starting at $\frac{L}{2}$ distance from the left end can be represented by Macaulay's function, where L is the length of the beam. Find the Macaulay's function for bending moment.
- (j) Define critical load in buckling. Compare the critical loads for two columns with fixed-free and fixed-fixed configurations.

(P.T.O.)

PART B

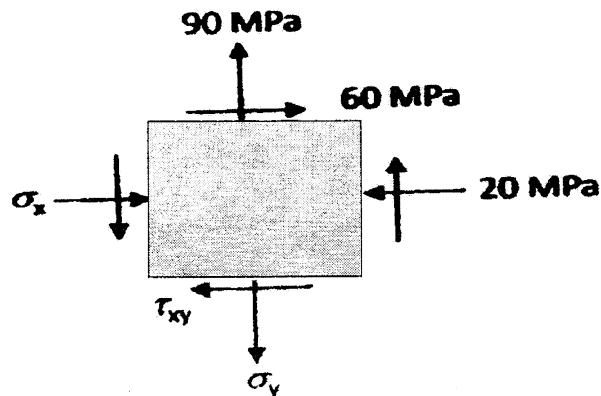
(4 × 10 = 40)

- II. An aluminium rod is rigidly attached between a steel rod and a bronze rod as shown in the figure below. Axial loads are applied at the positions indicated. Find the maximum value of P that will not exceed a stress in steel of 140 MPa, in aluminium of 90 MPa, or in bronze of 100 MPa.



OR

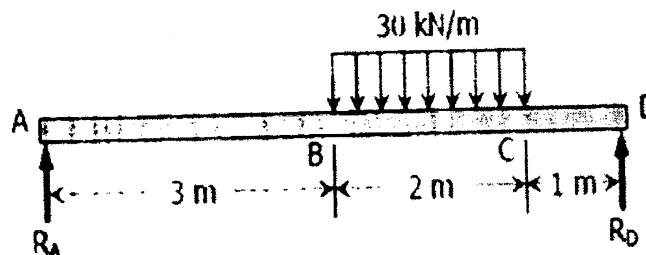
- III. The state of plane stress at a point on a body is shown on the element in the figure below. Represent this stress state in terms of the principal stresses.



- IV. A solid steel shaft 5 m long is stressed at 80 MPa when twisted through 4° . Using $G = 83 \text{ GPa}$, compute the shaft diameter. What power can be transmitted by the shaft at 20 Hz?

OR

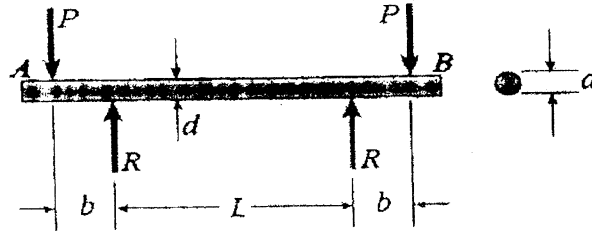
- V. A simply supported beam AD, 6 m long is shown below. A uniformly distributed load of 30 kN/m is applied as shown in the figure. Draw the shear-force and bending-moment diagrams for this beam.



(Contd...3)

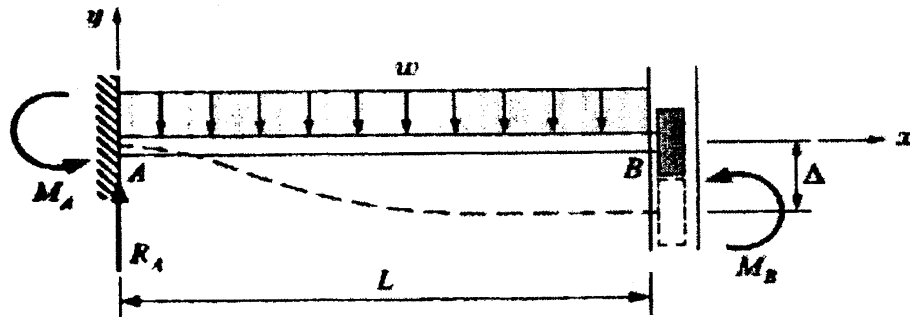


- VI. A freight-car axle AB is loaded approximately as shown in the figure, with the forces P representing the car loads (transmitted to the axle through the axle boxes) and the forces R representing the rail loads (transmitted to the axle through the wheels). The diameter of the axle is $d = 82 \text{ mm}$, the distance between centers of the rails is L , and the distance between the forces P and R is $b = 220 \text{ mm}$. Calculate the maximum bending stress $\sigma_{\max}$ in the axle if $P = 50 \text{ kN}$.



OR

- VII. Derive the Von Mises (equivalent) stress for planes stress ($\sigma_1, \sigma_2, \sigma_3 = 0$). Draw the stress ellipse.
- VIII. The horizontal beam AB shown in the figure below is clamped at A , subject to a uniformly distributed load, and supported at B in such a manner that it is free to deflect vertically but is completely restrained against rotation at that point. Determine the vertical deflection at B after the beam has deflected as shown by the dotted line.



OR

- IX. Determine the allowable axial load on a column with $I = 4.2 \times 10^6 \text{ mm}^4$ and $A = 3600 \text{ mm}^2$ and 4 m long. The ends are pinned, the yield point is 240 MPa and $E = 200 \text{ GPa}$.

